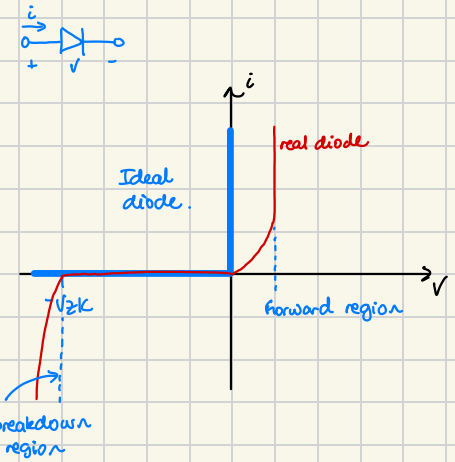
ELG 2136

Lecture 1

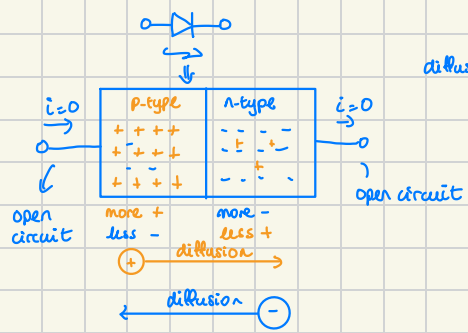
PN junctions



Silicon technology

add impurities:

- Phosphorus
- Boron



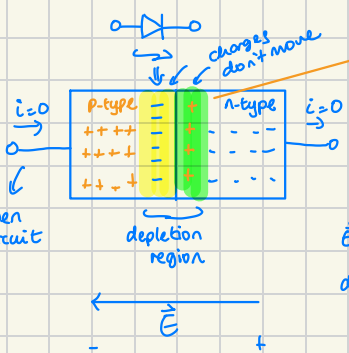
diffusion: more to less

Diffusion => flow of charges -> current flow

But we have an open-circuit on both sides -> current cannot flow [physically impossible]

-> Diffusion will STOP

Hence



Free from holes -> -ve charged

Free from electrons -> +ve charged

$\vec{E}$ helps minorities while ditching majorities.

drifting => movement due to $\vec{E}$ field.

Thermal voltage

$$|V_0| = V_T \ln\left(\frac{n_i^2}{N_A N_D}\right)$$

$$V_T = \frac{kT}{q} \doteq 26[mV]$$

n_i: intrinsic carrier concentration

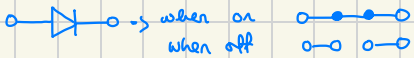
N_A: acceptor doping concentration
accept electrons, creating holes
[in p-type material]

N_D: donor doping concentration
donate electrons
[in n-type material]

k: Boltzmann Constant

T: temperature [K]

q: electron charge

Ideal diode: 

real diode mode: 